KYLE DAWSON, PHD

PhD Applied Scientist | Mathematical Optimization, Machine Learning & Generative AI

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EXPERIENCE

Lead Data Scientist — Energy

Centerpoint Energy

Apr 2026 – Present Houston, TX

- lead a new applied-science team, setting technical direction, mathematical-modeling and evaluation standards, and mentoring scientists and engineers across machine-learning and optimization initiatives
- designed and productionized a large-scale budget-constrained optimization model (mixed-integer linear programming via CBC with greedy heuristics, LightGBM risk scoring) that allocates a fixed operational budget to maximize grid reliability (SAIFI) per dollar — translating an ambiguous business problem into a rigorous OR formulation deployed on GCP
- built a deployed multi-LLM / generative-AI application (Gemini on Vertex AI, Cohere North) on Cloud Run with a retrieval benchmark scoring citation accuracy against a ground-truth set, to evaluate foundation models for document Q&A and image analysis
- ship production-grade Python systems (Docker, CI/CD, BigQuery) that take models from research prototype through simulation and experimentation to reliable, large-scale production

Senior Data Scientist — Climate Tech

SeekOps Inc.

Mar 2022 – Apr 2026 Austin, TX

- built and productionized machine-learning models to forecast survey quality, detect anomalies, and constrain uncertainty in gas-emission estimates, delivered through customer-facing analytics platforms (Plotly Dash on Azure) securing contracts of $50–150k
- implemented Bayesian uncertainty propagation and physics-based simulation models to generate decision-grade metrics used in regulatory and enterprise reporting across 10+ countries
- invented novel measurement and quantification methodologies, presenting results to executives, customers, and international peer-reviewed conferences

Data Scientist — Ag Tech

Bayer CropScience

Feb 2020 – Mar 2022 St. Louis, MO

- applied machine learning and statistical modeling to large-scale UAV-derived imagery for digital phenotyping and crop-performance assessment
- trained and evaluated deep-learning object-detection models (Detectron2) for classification from high-resolution aerial imagery
- built end-to-end ML data workflows combining relational databases, geospatial data, and AWS S3 into machine-learning-ready datasets serving stakeholder decision support

Postdoctoral Fellow

NASA Langley Research Center

Oct 2017 – Feb 2020 Hampton, VA

- developed uncertainty-aware retrieval algorithms from remote-sensing observations and delivered peer-reviewed results, prototyping and experimenting with models against field data

SELECTED PROJECT

BESS Day-Ahead Arbitrage Optimizer

Self-directed proof of concept

EXECUTIVE SUMMARY

PhD applied scientist with 8+ years inventing, experimenting with, and productionizing machine-learning, optimization, and simulation systems for scientifically complex, large-scale problems. Combines mathematical optimization (MILP/LP), machine learning, and generative AI to turn ambiguous business problems into rigorous models and shipped software. Peer-reviewed researcher who carries solutions from prototype and simulation through experimentation into reliable cloud production, and communicates results to technical and business stakeholders.

CORE COMPETENCIES

Mathematical Optimization · Operations Research · Research & Invention · Simulation & Experimentation · Research-to-Production · Model Evaluation & Validation · Technical Leadership · Mentoring & Team Building · Cross-Functional Collaboration · Stakeholder Communication

DOMAIN EXPERTISE

Mathematical Optimization (MILP/LP) · Machine Learning · Deep Learning · Forecasting & Time Series · Generative AI & LLMs · Anomaly Detection · Bayesian Inference · Uncertainty Quantification · Physics-Based Simulation · Computer Vision · Geospatial & Remote Sensing

TOOLS & TECHNOLOGIES

Python · PuLP / CBC · PyTorch · Scikit-Learn · LightGBM · XGBoost · Detectron2 · NumPy · Pandas · SQL · GCP BigQuery · Vertex AI · AWS · Azure · Docker · GPU / CUDA · MLOps / CI-CD · Git · Plotly · Anthropic Claude

AWARDS

NASA Postdoctoral Program Fellow

NASA Langley Research Center

Oct 2018 Hampton, VA

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Langley Aerospace Research Scholar

NASA Langley Research Center

Jun 2013 Hampton, VA

PUBLICATIONS

Google Scholar Profile

2025

A PuLP/CBC linear program, fed by ERCOT market API data, that schedules day-ahead battery charge/discharge arbitrage against nodal settlement-point (LMP) prices under state-of-charge and power constraints.

EDUCATION

PhD, Atmospheric Science

North Carolina State University

Graduated 2017 Raleigh, NC

Contributions to the Understanding of Aerosol Microphysics Towards Improving the Assessment of Climate Radiative Forcing